

Automatic Speech Translation

Laboratory Work 3: Fine-tuning Whisper for English→Ukrainian Translation on FLEURS

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1 Introduction

The goal of this lab is to build an Automatic Speech Translation (AST) system for the Ukrainian language. Unlike standard Automatic Speech Recognition (ASR), where the output stays in the same language as the audio, AST requires the model to take audio in one language and directly produce text in another. In this work the input is English audio and the expected output is Ukrainian text.

To implement this, we used OpenAI’s Whisper model, a general-purpose speech model trained on a large multilingual dataset. Whisper supports translation as a built-in task, which makes it a natural fit. The approach is to fine-tune the pretrained whisper-small checkpoint on a parallel English–Ukrainian speech dataset so it learns to perform this specific direction.

2 Dataset

2.1 Google FLEURS

The dataset is Google FLEURS (Few-shot Learning Evaluation of Universal Representations of Speech). It contains audio recordings of people reading Wikipedia-like sentences across many languages. Crucially, the same sentences are recorded in multiple languages, so it is possible to align English audio with Ukrainian transcriptions by matching sentence IDs. This gives us (source audio, target text) pairs needed for AST.

We loaded two subsets from HuggingFace: `en_us` and `uk_ua`. Samples were matched by the sentence `id` field and filtered to remove clips longer than 20 seconds.

Table 1: Dataset split statistics after filtering and parallel alignment.

Split	Samples	Duration (h)	Avg clip (s)
Train	2,465	6.88	~10.1
Validation	357	0.92	~9.3
Test	601	1.62	~9.7

2.2 Preprocessing

Audio was resampled to 16 kHz. Text was normalised by stripping whitespace and standardising apostrophe variants. Whisper’s feature extractor converts raw waveforms into 80-channel log-Mel spectrograms padded or truncated to 30 seconds (3,000 time steps). Labels are tokenised with Whisper’s tokeniser and padded inside a custom collator, with pad positions set to -100 so the cross-entropy loss ignores them.

3 Model and Training Setup

3.1 Model Architecture

The base model is `openai/whisper-small`, an encoder-decoder Transformer with 241 million parameters. The encoder is a convolutional audio frontend followed by 12 Transformer layers (hidden size 768). The decoder is also 12 layers and generates text autoregressively with a vocabulary of 51,865 tokens.

The encoder was frozen during training, so only the decoder’s 153 million parameters were updated. The encoder already has strong audio representations from Whisper’s pre-training, so freezing it reduces compute cost without hurting the feature extraction. The decoder is where the translation behaviour needs to be learned.

Forced decoder IDs were set to `ukrainian + translate` task, which is how Whisper is controlled for language and task selection at inference time.

3.2 Training Configuration

Table 2: Training hyperparameters.

Hyperparameter	Value
Base model	openai/whisper-small
Trainable parameters	153M (decoder only; encoder frozen)
Dataset	Google FLEURS (en_us $\rightarrow$ uk_ua)
Train / Val / Test	2,465 / 357 / 601 samples
Learning rate	5×10^{-6} with linear warmup (5%)
Weight decay	1×10^{-2}
Effective batch size	2 per GPU $\times$ 8 grad. accumulation = 16
Max epochs	5 (best checkpoint: epoch 4)
Precision	16-bit mixed (AMP)
Max new tokens	128
Decoding	Greedy (num_beams = 1)
Gradient clipping	1.0

Training was run on a Tesla T4 GPU using PyTorch Lightning. Mixed precision (FP16) was used to save memory. The effective batch size is 16 (per-device batch size 2 with gradient accumulation of 8 steps). Early stopping monitored validation chrF with patience of 2 epochs. The best checkpoint was saved at epoch 4 with `val_chrf` = 26.188.

4 Evaluation Metrics

Three metrics were used to evaluate translation quality.

- **BLEU** measures n-gram overlap between prediction and reference. It is the standard metric in machine translation but is sensitive to exact word matches. Ranges from 0 to 1.
- **chrF** is a character-level F-score, more robust for morphologically rich languages like Ukrainian where the same word can appear in many inflected forms. Ranges from 0 to 100.
- **COMET** (wmt22-comet-da) is a neural metric that scores translations using a pretrained model conditioned on source, prediction, and reference together. Ranges roughly from 0 to 1.

All metrics were computed on the held-out test set after loading the best checkpoint.

5 Results

5.1 Test Set Scores

Table 3 summarises the final scores on the test set of 601 samples.

Table 3: Final test set results.

Metric	Score	Scale	Notes
BLEU	0.0199	0–1	Low; exact match sensitive
chrF	26.11	0–100	Char-level; fairer for Ukrainian
COMET	0.4101	0–1	Neural quality metric

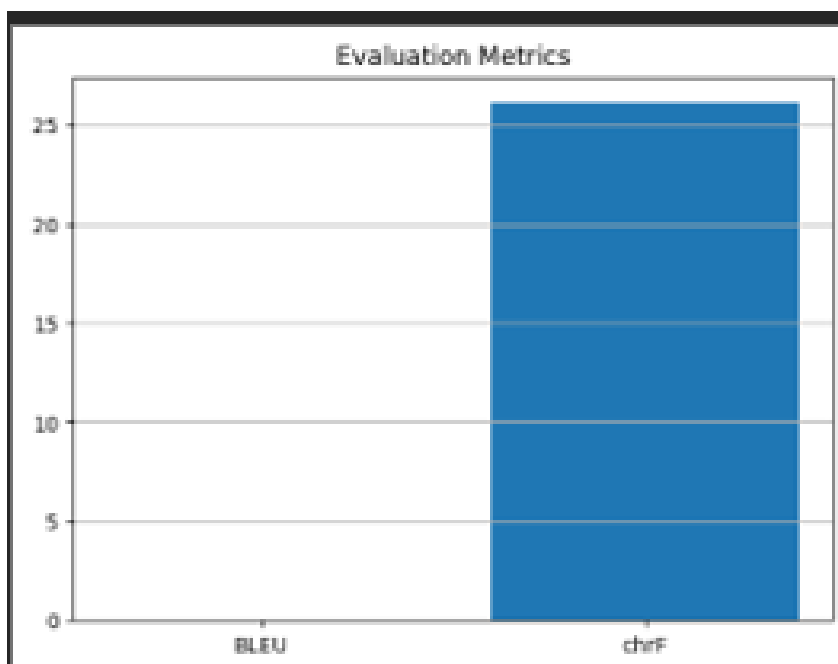


Figure 1: BLEU and chrF scores on the test set.

BLEU appears near-zero on this scale because it is orders of magnitude smaller than chrF. The BLEU score of 0.0199 is low, but expected. BLEU requires exact n-gram matches, and Ukrainian is a highly inflected language where correct translations often differ in word endings. A translation can be semantically correct and still score near zero on BLEU.

The chrF score of 26.11 gives a fairer picture because character-level matching is more tolerant of morphological variation. The COMET score of 0.41 from a neural metric confirms the model is producing meaningful output, not random tokens.

5.2 Prediction Length Analysis

Figures 2 and 3 compare the character lengths of predictions and references on the test set.

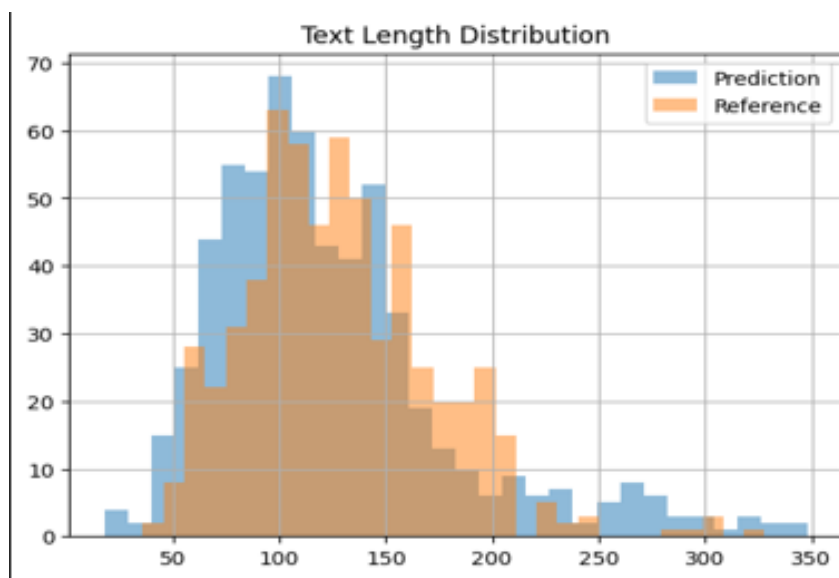


Figure 2: Distribution of output text lengths (characters) for predictions vs. references.

The distributions are similar in shape, both centred around 100–150 characters. Predictions are slightly shorter on average, suggesting some content is occasionally lost. The tails contain a few very long predictions which may indicate hallucination.

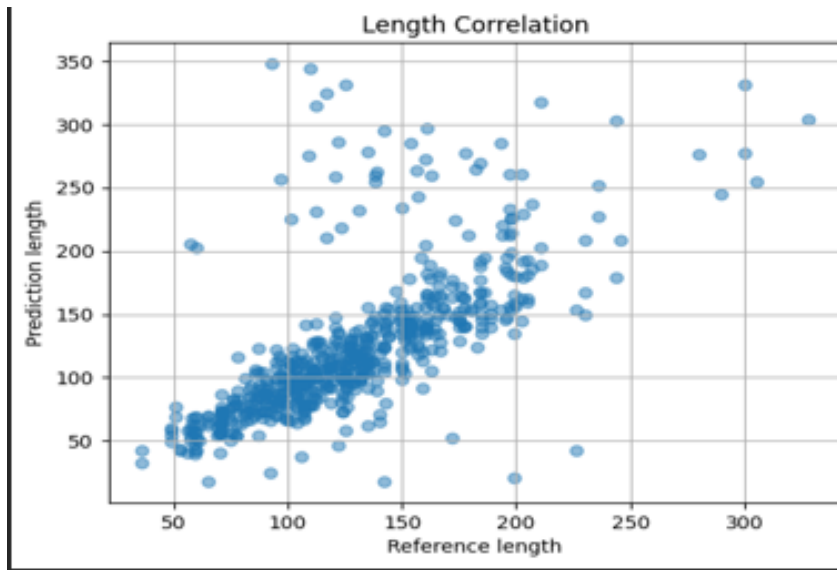


Figure 3: Scatter plot of reference length vs. prediction length per test sample.

Points near the diagonal indicate the output length is correct. The scatter plot shows a positive correlation between reference and prediction lengths, meaning the model broadly understands how much to generate. However, scatter is notable for longer references, and some short references get unexpectedly long predictions.

5.3 Sample Predictions

Table 4 shows three qualitative examples from the test set.

Table 4: Sample reference–prediction pairs from the test set.

Reference	Prediction
всі живуть у суспільстві і користуються транспортними системами при цьому мало не всі скаржаться на них	все вирішують в соціальному і використовують транспортні системи майже всі варіють про транспортні системи
ви також можете захотіти проконсультуватися з урядами інших країн окрім вашої але їхні поради розроблені для їхніх громадян	можливо ви також хочете послугувати влади іншими своїми але своє послугу є зробити для своїх жителів
атомна бомба працює за таким принципом що вона бере енергію щоб зібрати разом ядро з багатьма протонами та нейтронами	фізичний бом був на принципі того що виглядає енергії щоб збираючися на кукульту з багатим протоном та нейтроном

The model consistently produces Ukrainian text and captures the general topic. The main errors are wrong inflectional endings, incorrect word forms, and occasionally shifted

meaning. Predictions are grammatically plausible but not reliable. This is consistent with the numeric metrics.

6 Discussion

6.1 What Worked

Freezing the encoder was a reasonable decision given compute constraints. The decoder learned to produce Ukrainian output without disrupting the audio feature extraction. Training converged over 5 epochs and the model consistently generates valid Ukrainian sentences rather than garbled tokens.

6.2 Limitations

The training set is relatively small — only about 6.9 hours of audio. For a cross-lingual translation task where the model must learn to both understand English speech and generate Ukrainian text, this is quite limited. Greedy decoding (`num_beams = 1`) is fast but suboptimal for translation quality — beam search would likely improve all three metrics. The `max_new_tokens = 128` limit may also truncate longer outputs.

6.3 Possible Improvements

Several directions could improve results. Using a larger Whisper variant (medium or large-v2) would increase model capacity. Partially unfreezing the encoder with a very small learning rate could help it better adapt to this translation direction. Adding more Ukrainian–English parallel data and using beam search decoding would be the most impactful changes.

7 Conclusion

We fine-tuned `openai/whisper-small` on Google FLEURS for English-to-Ukrainian speech translation using PyTorch Lightning and obtained $\text{BLEU} = 0.0199$, $\text{chrF} = 26.11$, and $\text{COMET} = 0.41$ on the held-out test set. The pipeline covers parallel data construction by sentence-ID matching, a custom dataset and collator, mixed-precision training with early stopping and checkpointing, and evaluation with three complementary metrics. The best checkpoint was saved at epoch 4. The model successfully generates Ukrainian text from English audio, confirming the approach works. The main bottlenecks are limited training data and greedy decoding.